Optics Communications 582 (2025) 131654

Contents lists available at ScienceDirect

Optics Communications

journal homepage: www.elsevier.com/locate/optcom

Machine learning-assisted prediction of organic solar cell efficiency from TCA triplelayer reflectance spectra

Fuhao Gao [a], Jinxin Zhou [a], Junwei Zhao [a], Senxuan Lin [a], Jingfeng Liu [a], Yubin Lan [a,b], Yongbing Long [a,b,c,*], Haitao Xu [a,**]

[a] College of Electronic Engineering/College of Artificial Intelligence, South China Agricultural University, Guangzhou, 510642, PR China
[b] Lingnan Modern Agr Sci & Technol Guangdong Lab, Guangzhou, 510642, PR China
[c] South China Agricultural University, Natl Ctr Int Collaborate Res Precis Agr Aviat Pest, Guangzhou, 510642, PR China

ARTICLE INFO

Keywords:
Organic solar cells
Machine learning
Spectroscopic analysis
PCA
CARS
PCE

ABSTRACT

Organic Solar Cells (OSCs) are one of the most promising solar cells due to the possible for large-scale and low-cost printed production. Therefore, efficient manufacturing processes and optimization methods are crucial. Currently, the traditional trial-and-error method is mostly used to optimize device performance, which is complex and time-consuming. Previous machine learning (ML) methods can reduce the workload, but relay on multiple inputs. To accelerate the optimization process, a novel ML-based approach was proposed to predict the power conversion efficiency (PCE) of OSCs, utilizing the reflectance spectrum of the transparent electrode/charge transport layer/active layer (TCA triplelayer). For this purpose, a dataset, containing PCEs of six types of OSCs with different active layer materials and reflectance spectra of TCA triplelayers, had been constructed by simulations via finite-difference time-Domain method. Based on the dataset, machine learning algorithms were employed to construct the regression models. Spectra pre-processing and feature extraction techniques were integrated to refine the predictive accuracy of these models. Consequently, the model based on Multilayer Perceptron Regression (MLPR) algorithm demonstrated the best performance, with coefficient of determination (R^2) of 0.984 and root-mean-squared error of 0.408. These results underscore the potential to accurately predict the PCE of OSCs from the reflectance spectra of TCA triplelayer. Ultimately, a strategy was further proposed to utilize the developed regression model for real-time quality monitoring of TCA triplelayer during device fabrication. This offers a rapid way to evaluate the quality of TCA triplelayers and their influence on device performance.

1. Introduction

Organic solar cells (OSCs) have emerged as a focal point of research due to the flexibility, the capability for large-scale and low-cost printed production, and their potential for alleviating severe energy and environmental issues [1–3]. Consequently, a vast amount of research has been devoted to enhancing the performance of OSCs, including the power conversion efficiency (PCE), which has reached a notable value of over 20% [4].

The preparation process of OSCs usually involves layer-by-layer fabrication of multiple layers: spin-coating charge transfer layer and active layer on indium tin oxide (ITO) transparent electrode (these three layers are referred to as TCA triplelayers in this paper) and then depositing another charge transfer layer and metal back electrode on the TCA triplelayers. To obtain OSC devices with maximum efficiency, hundreds or even thousands of OSCs need to be prepared to optimize device structure and J-V curves of all these OSCs should be measured to obtain the PCE of devices [5,6]. The optimization process is inherently time-consuming, labor-intensive and complicated. In particular, the deposition of second charge transfer layer and metal back electrode is very troublesome since the processing is usually performed in a vacuum environment. Therefore, it is critical to develop a real-time evaluation method for device efficiency after the TCA triplelayers have been fabricated since such an evaluation method can help to reduce the workload by predicting the PCE of the entire device and eliminating the need for subsequent preparation steps such as charge transfer layer and

* Corresponding author. College of Electronic Engineering/College of Artificial Intelligence, South China Agricultural University, Guangzhou, 510642, PR China
** Corresponding author.
E-mail address: xuhaitao@scau.edu.cn (H. Xu).

https://doi.org/10.1016/j.optcom.2025.131654
Received 18 December 2024; Received in revised form 5 February 2025; Accepted 20 February 2025
Available online 21 February 2025

back electrode deposition. In recent years, machine learning (ML) technology has been extensively employed in the material screening, key feature identification, performance prediction, morphological analysis, process optimization and high-throughput automation platforms to facilitate the advancement of OSCs [7,8]. In terms of material screening, Irfan et al. successfully constructed an ML model with electrical properties as input to select small molecule donor materials suitable for Y6 [9]. Several novel small-molecule donor materials were designed based on the predictions of the ML models. In the aspect of identification of key feature, Huang et al. developed an ML model for quantitative analysis of the relationships between non-radiative voltage loss and the structure, electronic properties of donors and acceptors in OSCs [10]. By employing the optimal ML model, the researchers have predicted that Y18B derivatives possess the smallest HOMO-LUMO gap and optical band gap, thus contributing to the effective reduction of voltage loss within OSCs. In the terms of performance prediction, Janjua et al. employed the ML method to investigate the correlation between electronic parameters and the PCE of OSCs, and used these parameters to predict the PCE [11]. Malhotra et al. introduced a novel directed message passing neural network (D-MPNN), which was specifically designed to predict the PCE of OSCs [12]. Notably, the D-MPNN exhibited superior predictive performance in comparison to other traditional ML models. In the field of morphological analysis, Cui et al. used ML to optimize the morphology parameters and electrical properties of OSCs, and realized the simultaneous improvement of PCE and thermal stability [13]. In the application of process optimization, Wang et al. investigated the correlation between solar cell performance and the processing parameters of all-small molecule heterojunction solar cells by ML, and fabricated solar cells, achieving a quite high efficiency of approximately 4% according to the processing parameters

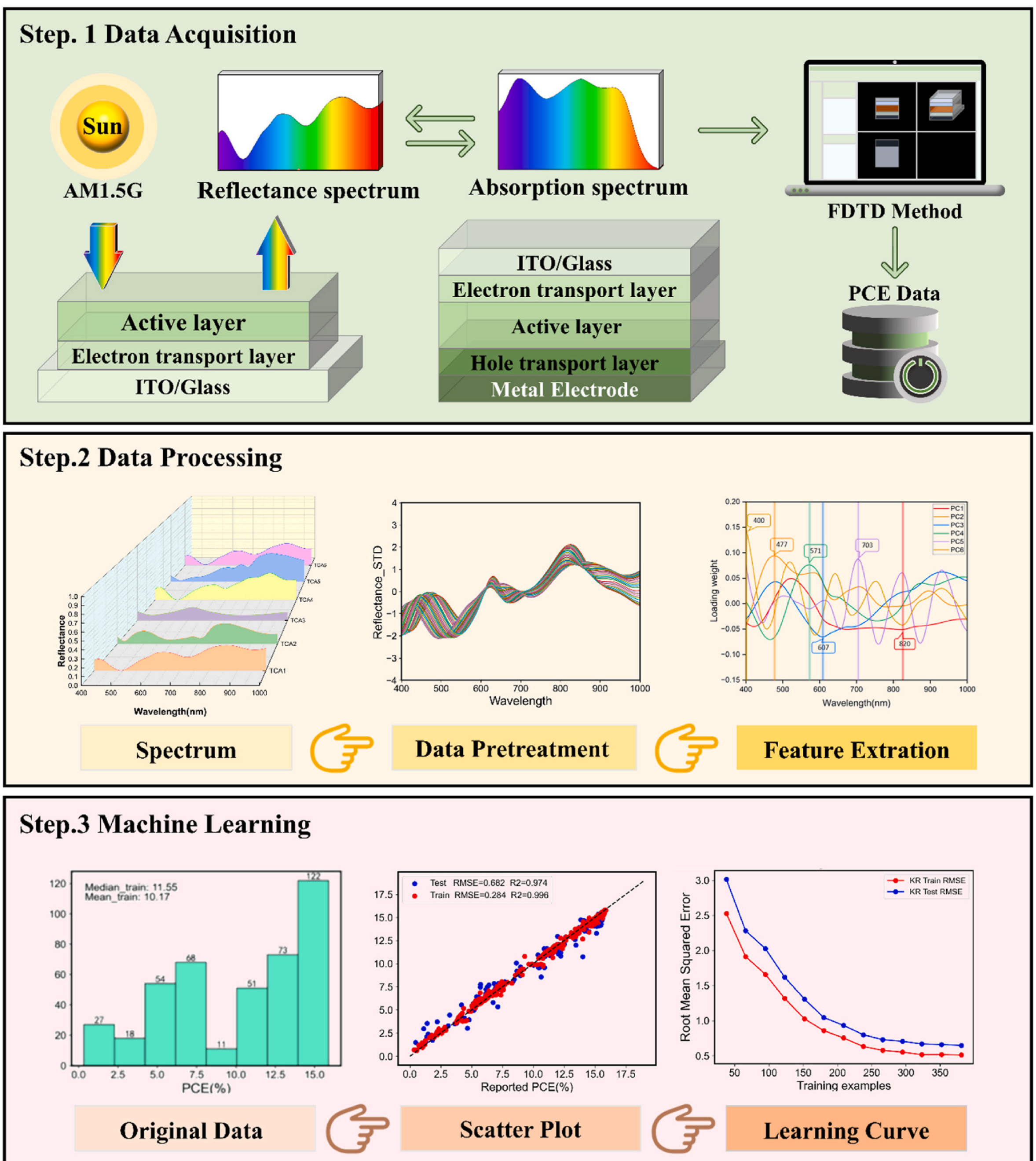

Fig. 1. Workflow of constructing models to predict the PCE of OSCs.

recommended by ML [14]. Finally, in the application of high-throughput automation platforms, Du et al. used ML to assess the link between processing stability and chemical, energetic, and morphological structures, and successfully found that a similarly good prediction of air/light resilience can also be achieved by considering only the features in the chemical structure, that is, the information available before any experiment [15].

Although previous ML-based approaches have provided guidance for designing and optimizing the performance of OSCs, these approaches are not suitable for real-timely evaluating device efficiency by evaluating the quality of the TCA triplelayers. Fortunately, the spectral analyzing combined with ML models has the advantages for real-time quality monitoring [16,17]. The reflectance spectra usually bring much information of the materials and structure of the illuminated objects, enabling rapid and nondestructive measurement of internal components and quality of objects [18,61]. This indicates that the use of reflectance spectra combined with the ML models may provide a new approach to evaluate the device efficiency via the reflectance spectra of the TCA triplelayers, which is not yet investigated in previous studies.

These issues were investigated in paper by developing a ML-assisted PCE prediction method for OSCs based on the reflectance spectra of TCA triplelayer. The specific objectives of the paper are as follows: (1) construct a simulated dataset for training the PCE prediction model, which comprises the PCEs of six types of OSCs and the reflectance spectra of the corresponding TCA triplelayers; (2) analyze the relationship between reflectance spectra of TCA triplelayer and PCE of the device; (3) construct ML models to predict PCE of the devices from the reflectance spectra of TCA triplelayers; (4) propose an novel approach based on the ML model to evaluate the quality of the TCA triplelayer by predicting the PCE of the OSC device after the active layer is spin-coated. The main workflow for constructing prediction models for the PCE of OSCs is illustrated in Fig. 1.

2. Materials and methods

2.1. Device structure

The structures of the six types of OSCs, together with the corresponding TCA triplelayer were summarized in Table 1. Each OSC device had a structure of adding two further layers, namely a hole transport layer and an Ag back electrode, on the top of TCA triplelayer. In both OSCs and TCA triplelayer, various donor and acceptor materials were used as the component of the active layer (PM6:Y6, PTB7:PC_{71}BM, P3HT:PCBM, PM6:BTP-ec9 and PBDB-T: ITIC), which includes polymer donors (PM6, PTB7, P3HT, PBDB-T), small molecule acceptors (Y6, ITIC) and the polymer acceptors (PC_{71}BM, PCBM, BTP-ec9). ITO film with thickness of 150 nm and Ag film with thickness of 100 nm were used as transparent electrode and back electrode, respectively. Zinc oxide (ZnO) and PDINN with thickness of 30 nm were used as electron transport layer in the OSCs. The PEDOT:PSS and molybdenum trioxide (MoO_3) with thickness of 10 nm were used as hole transport layer. The OSC devices were referred to as OSC 1, OSC 2, OSC 3, OSC 4, OSC 5, and OSC 6. The corresponding TCA triplelayer structures were labeled as TCA 1, TCA 2, TCA 3, TCA 4, TCA 5, and TCA 6, respectively. For OSC 1 and TCA 1, the thickness of the active layer is assumed to be varied from 50 nm to 300 nm with an interval 2 nm. For other devices and TCA triplelayers, the thickness of the active layer was varied from 10 nm to 200 nm with an interval 2 nm. As a result, 606 OSC devices together with the TCA triplelayers were achieved to construct the PCE prediction models. All refractive indexes used in the calculations were sourced from the literatures [19–21].

2.2. Dataset construction

The dataset used to train the PCE prediction models was composed of the PCEs of 606 OSC devices and the reflectance spectra of the corresponding TCA triplelayers, of which the structure and parameters was presented Section 2.1. The PCE and reflectance spectra were obtained by employing Finite-Difference Time-Domain (FDTD) method and more details are as follows.

The reflectance spectra of the TCA triplelayer are calculated by dividing the optical electric field intensity of the reflected and incident waves, as represented by the following equation [22]:

$$R = \frac{|E\_refl|^2}{|E\_Inc|^2} \tag{1}$$

where $|E\_refl|^2$ and $|E\_Inc|^2$ represent the square of the modulus of optical electric field for the reflected wave and the incident wave.

To calculate the PCE of the OSCs, FDTD was first employed to calculate the absorption spectra $A(\lambda)$ of the active layer within the device, which can be directly obtained by a monitor. By assuming all photons absorbed in the active layer can be collected by the electrodes, the short-circuit current density (J_{sc}) of the devices can be calculated as [23]:

$$J_{sc} = \frac{q}{hc} \times \int A(\lambda) \times P_{AM1.5G}(\lambda) \times \lambda d\lambda \tag{2}$$

where q is the electron charge, h is Planck's constant, and c is the speed of light in a vacuum, λ represents the wavelength of the incident light.

The PCE of OSCs can be calculated using formula [24]:

$$PCE = \frac{V_{OC} \times J_{SC} \times FF}{P_{in}} \tag{3}$$

where P_{in} is the input power, FF is the fill factor, and V_{oc} is the open-circuit voltage. The values for FF and V_{oc} were collected from previous references [25–29]. The V_{oc} and FF values reported in the literature represent the comprehensive results obtained by experimental researchers after taking into account multiple key factors, such as the band gap of the materials, the morphology of the active layer, non-radiative recombination, and the device structure. In other words, although the J_{sc} is obtained by the FDTD method, the PCE here takes into account, to some extent, the influencing factors in reality. All other conditions remain constant, and the PCE calculated by the FDTD method is shown in Table 2 for details. From Tables 2 and it can be found that the calculated PCE_{Cal} of OSC1, OSC2, OSC3, OSC4, OSC5 and OSC6 was about 0.64%, 0.11%, 0.54%, 0.66%, 1.2% and 0.78% higher than the experimental values (PCE_{Exp}) reported in the literatures, respectively [25–29]. The PCE difference occurs because PCE_{Cal} is achieved in an ideal condition such as perfect interfaces, uniform material properties

Table 1
The structural information of six types of OSCs and TCA triplelayer Structures.

Device	OSCs Structures	TCA triplelayer Structures
OSC 1	ITO/ZnO/PM6:Y6/MoO_3/Ag	ITO/ZnO/PM6:Y6
OSC 2	ITO/ZnO/PTB7:PC_{71}BM/MoO_3/Ag	ITO/ZnO/PTB7:PC_{71}BM
OSC 3	ITO/ZnO/P3HT:PCBM/MoO_3/Ag	ITO/ZnO/P3HT:PCBM
OSC 4	ITO/ZnO/PM6:BTP-ec9/MoO_3/Ag	ITO/ZnO/PM6:BTP-ec9
OSC 5	ITO/PDINN/PBDB-T:ITIC/PEDOT:PSS/Ag	ITO/PDINN/PBDB-T:ITIC
OSC 6	ITO/PDINN/PM6:Y6/PEDOT:PSS/Ag	ITO/PDINN/PM6:Y6

Table 2
The PCE calculated by FDTD method.

Device	Thickness of active layer(nm)	PCE_{Cal}(%)	PCE_{Exp}(%)	References
OSC 1	100	15.54	14.90	[25]
OSC 2	100	3.37	3.26	[26]
OSC 3	100	4.18	3.64	[27]
OSC 4	140	15.67	15.01	[28]
OSC 5	100	10.73	9.53	[29]
OSC 6	100	15.78	15.00	[29]

and assumptions that all photogenerated electrons contribute to current. Another reason for the PCE difference is that the refractive index of the materials may be different for the OSC devices in these literatures due to the different material preparation conditions. Nonetheless, PCE_{Cal} obtained by FDTD is in an acceptable error range when compared with experimental results.

2.3. Data pre-processing

Spectral data pre-processing is an important step to improve the prediction accuracy of qualitative models, as it can mitigate adverse effects and improve the quality and reliability of the reflectance spectra [30]. For this purpose, four methods are employed to preprocess the reflectance spectra within the dataset. These include the Savitzky-Golay smoothing filter (SG), the Standardization (STD), the First derivative (D1), and the Second derivative(D2).

SG is a filtering method based on local polynomial least squares fitting in the time domain, and its basic principle is to obtain smoothed data points by polynomial fitting of data points in a sliding window [31]. SG is an effective method for reducing noise while maintaining the shape and width of the signal.

STD is a common data pre-processing method that operates by subtracting the mean from each data point and dividing by the standard deviation. After this processing, the data will conform to a standard normal distribution, characterized by a mean of 0 and a standard deviation of 1. Due to the different active layer materials, there may be numerical differences in the reflectance spectra of TCA triplelayer. The STD can be employed to transform these spectral data onto a uniform scale, specifically a distribution with a mean of 0 and a standard deviation of 1. This process eliminates differences caused by the material properties of the various active layer materials, thereby enhancing the predictive ability of the model [32].

D1 and D2 represent the slope of the reflectance spectra data at a specific point and the second derivative of the raw data, respectively. D1 is essential for identifying peaks and dips in the reflectance spectrum [33], as these extreme points are often closely related to the optical properties of the material. D2 is derived by differentiating D1, and can also be utilized to detect peaks and dips in spectrum. In comparison to D1, D2 is more effective at distinguishing between overlapping or similar peaks. When two TCA triplelayer exhibit similar or overlapping reflectance, D2 can effectively distinguish the overlapping peaks [34].

2.4. Feature extraction

In reflectance spectra, adjacent wavelengths may contain similar information, which in turn gives rise to redundancy in the data and a consequent reduction in the accuracy of the prediction model. The application of feature extraction techniques is essential, as they not only remove superfluous characteristics but also enhance the prediction accuracy of the model. In this study, Principal Component Analysis (PCA) [35] and Competitive Adaptive Reweighted Sampling (CARS) [36] were utilized for feature extraction.

PCA is a statistical methodology that employs an orthogonal transformation to convert potentially linearly correlated high-dimensional original features into a set of linearly uncorrelated principal components. The components are then ordered according to the size of their variance, thereby preserving the principal variability information within the data. This approach has the potential to significantly reduce the dimensionality of the data while endeavoring to retain key information from the original data. In this study, the proportion of variance in the original data explained by each principal component was calculated and these proportions were accumulated until a predetermined threshold of 90% was reached. This approach prevents the potential information loss that could result from selecting too few principal components, while also avoiding the overfitting issues that could arise from choosing an excessive number of components.

The CARS algorithm is based on the principle of "survival of the fittest", as observed in evolutionary biology. It is a feature variable selection method that combines Monte Carlo sampling with regression coefficients derived from the Partial Least Squares (PLS) model. In the CARS algorithm, each iteration involves adaptive reweighted sampling. During each iteration, wavelength points are sampled based on the current weights, a model is constructed and its performance is evaluated, and then the weights are updated based on the evaluation results. In the process of weight updating, the weights of poorly performing wavelength points are reduced, whereas the weights of those demonstrating superior performance are increased. After multiple calculations, the subset of wavelengths with the smallest Root Mean Square Error of Cross-Validation ($RMSE_{CV}$) in the PLS model is selected as the characteristic wavelengths.

2.5. Prediction models

A stable and reliable ML model can make accurate and reasonable predictions. Considering the complexity involved in predicting the PCE of OSCs from the reflectance spectra of TCA triplelayer structures, six ML algorithms have been tentatively selected for capturing data characteristics and constructing prediction models for the PCE of OSCs. Specifically, the employed algorithms include Partial Least Squares Regression (PLSR), K-Nearest Neighbors Regression (KR), Random Forest Regression (RFR), Support Vector Regression (SVR), Extreme Learning Machine (ELM), and Multilayer Perceptron Regression (MLPR).

PLSR is a multivariate analytical technique that projects the feature matrix of independent variables (X) and the dependent variable (Y) into the principal component space to explore linear relationships between X and Y. This method can be effectively used to investigate the linear relationship between the reflectance spectra of TCA and the PCE of OSCs [37].

KR is a non-parametric regression technique that predicts the label value of a test sample based on the distances between the test sample and the training samples [38]. During the prediction phase, the KR algorithm identifies the nearest neighbors (K) within the training set to the test sample and estimates the PCE of the test sample based on the known PCE values of these neighbors.

RFR is an ensemble learning technique that constructs multiple decision trees, each of which is trained on a randomly selected subset of features [39]. The final prediction outcome is the average of the predictions made by all trees, thereby establishing that the PCE estimated by RFR is the average value across numerous decision trees.

SVR is an approach derived from Support Vector Machine (SVM) for regression tasks, which operates by seeking an optimal hyperplane within the feature space to approximate the data closely [40].

ELM is a type of feedforward neural network that randomly assigns the weights and biases of the hidden layer nodes without requiring iterative training. The absence of iterative processes in ELM allows for rapid learning [41].

MLPR is a type of multilayer feedforward neural network that learns the mapping between input and output data through the use of at least one hidden layer. Each layer of the network consists of multiple neurons interconnected through weights [42]. Compared to ELM,MLPR iteratively adjusts the weights and biases within the network through backpropagation to minimize predictive error. The sophisticated architecture of MLPR enables it to effectively learn the complex non-linear relationships between the reflectance spectra of the TCA triplelayer and the PCE of OSCs.

2.6. Assessment of prediction models

To better evaluate the predictive performance of each model, R-Square (R^2) and Root Mean Square Error (RMSE) are used as performance metrics. R^2 and RMSE are expressed as the following equations:

$$R^2 = 1 - \frac{\sum_{i=1}^{n}(y_i - x_i)^2}{\sum_{i=1}^{n}(\bar{y} - x_i)^2} \tag{4}$$

$$RMSE = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(y_i - x_i)^2} \tag{5}$$

Here, x_i represents the true value of PCE in the dataset, y_i denotes the result predicted by the ML model, and $\bar{y}$ represents the mean of the predicted data. R^2 is primarily used to compare the fitting effects of different models. The closer the R^2 value is to 1, the better the fitting effect of the model. RMSE can be calculated for both the training set and the test set. The purpose of this is to check whether the model suffers from overfitting.

3. Results and discussion

3.1. Analysis of the relationship between reflectance spectrum and PCE

To analyze the relationship between reflectance spectra of the TCA triplelayer and PCE of OSC, we first compared the reflectance spectra of the TCA triplelayer and the absorption spectra of the corresponding OSC devices. Fig. 2(a) and (b) demonstrates that reflectance spectra of TCA 1 and absorption spectra of the corresponding OSC1 with active layer thickness set as 100 nm 120 nm,140 nm and 200 nm. As can be observed in Fig. 2(a), the reflectance spectrum showed oscillating behavior due to optical interference occurring in the TCA triplelayer, which also causes a red-shift of the spectral peaks [43,44]. In the TCA triplelayer, light is reflected at the interface of each layer due to the different refractive index of the transparent electrodes, charge transport layers, and active layers. When the thickness of the active layer changes, it alters optical interference pattern of the optical electric field, resulting in a variation in the reflectance spectra of the TCA triplelayer. Moreover, at the wavelength range (approximately 400–450 nm and 600–650 nm) where the reflectance shows peaks, the absorption spectra of the OSC device showed absorption valleys, as is shown in Fig. 2(b). Similar results can be also observed for other types of TCA triplelayer and OSCs, as is illustrated in Fig. 2 (c) and **(d)**. These findings indicate that there is a complicated relationship between the reflectance spectrum of the TCA triplelayer and the absorption spectrum of the corresponding OSCs.

The relationship between the reflectance spectrum of the TCA triplelayer and the absorption spectrum of the corresponding OSCs are further analyzed by employing thin optics [45–47]. Actually, the complex refractive index $\bar{n}(\lambda)$ of thin layers in TCA triplelayer can be described by the formula [48]: $\bar{n}(\lambda) = n(\lambda) + ik(\lambda)$, where $n(\lambda)$ is the refractive index, which represents the optical refractive properties of the material, $k(\lambda)$ is the extinction coefficient, which is related to the absorption coefficient of the material, λ denotes the wavelength of the incident light.

According to thin optics, the transport matrix of thin films in multilayer can be expressed as: $\begin{bmatrix} m_{11} & m_{12} \\ m_{21} & m_{22} \end{bmatrix} = \prod_{j=1}^{m} \begin{bmatrix} \cos\delta_j & \frac{i}{n_j}\sin\delta_j \\ in_j \sin\delta_j & \cos\delta_j \end{bmatrix}$, where $\delta_j = \frac{2\pi}{\lambda} n_j d_j$, d_j is the thickness of thin film [46]. And then, the reflectance spectra of the TCA triplelayer can be expressed by the formula [46]:

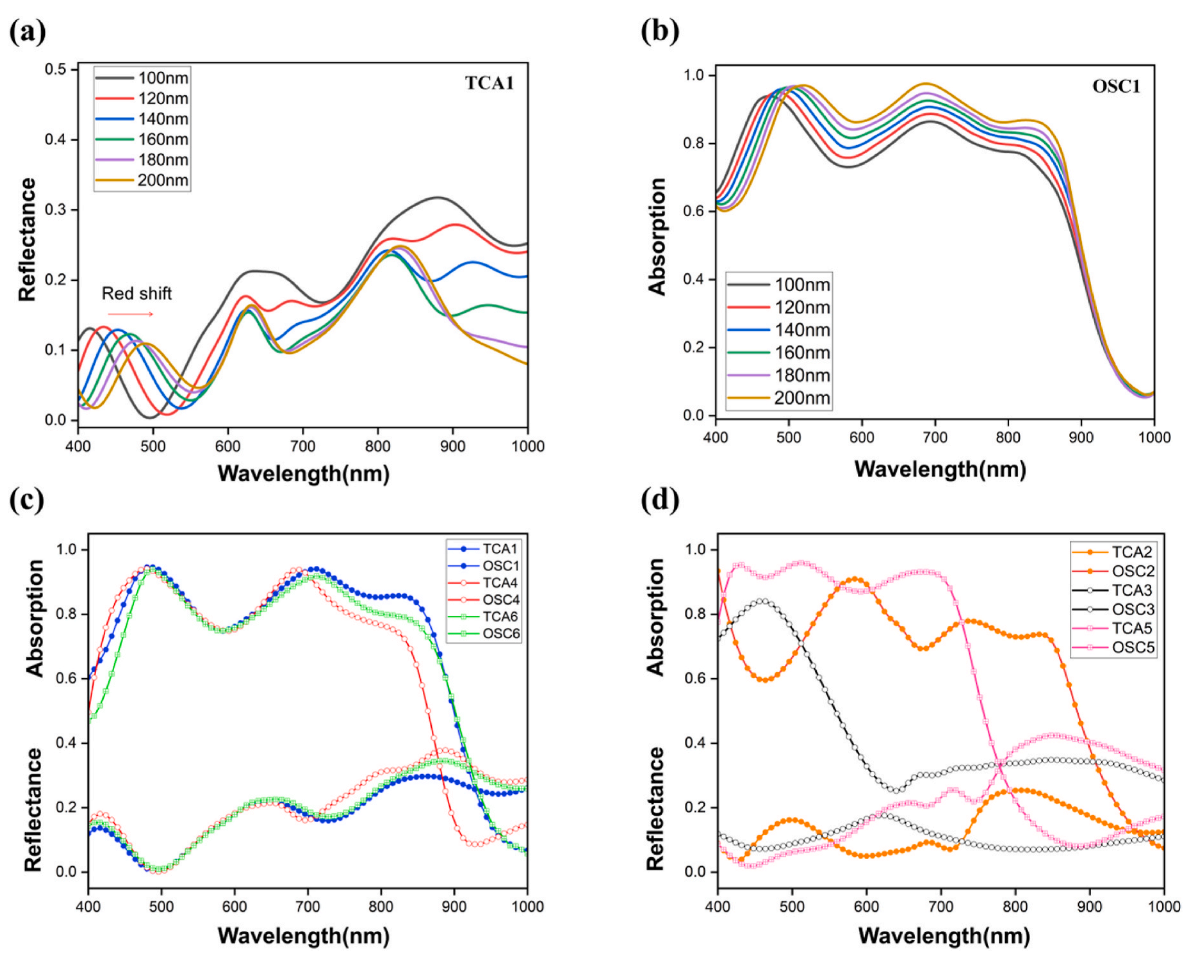

Fig. 2. (A) Reflectance spectra of TCA1 (b) Absorption spectra corresponding to OSC1 with different active layer thickness. (c) and (d) Reflectance spectra of TCA triplelayers and absorption spectra for corresponding OSC devices with the active layer thickness of 100 nm.

$$R = \frac{(n_o m_{11} - m_{22} n_{m+1})^2 + (n_o n_{m+1} m_{12} - m_{21} n_{m+1})^2}{(n_o m_{11} - m_{22} n_{m+1})^2 + (n_o n_{m+1} m_{12} - m_{21} n_{m+1})^2} \tag{6}$$

For simplification, R can be expressed as a function of the complex refractive index and thickness (d) of layers within structure:

$$R = f(\overline{n}(\lambda), d, \lambda) \tag{7}$$

For a TCA structure with a given reflectance spectrum, the thickness of the active layer can be calculated as an inverse function of $d = f^{-1}(R, \lambda)$. Subsequently, the absorption of the active layer in the corresponding OSCs can be expressed as follows:

$$A(d, \lambda) = g\left(f^{-1}(R, \lambda), \lambda\right) \tag{8}$$

The functions f and g within Equations (7) and (8) can be obtained according to the thin optics, which cannot be expressed as a simple expression.

According to Equations (2), (3) and (8), it is found that PCE of OSC devices have a direct relationship to the reflectance spectrum of the TCA triplelayer within devices. The relationship, however, is complicated since the relation between the absorption $A(d, \lambda)$ of active layer and the reflectance $f(d, \lambda)$ of TCA tripelayer is complicated. Fortunately, the complicated relationship can be learned by the machine learning method, which will be discussed in this paper. Schematic diagram for relationship between the PCE and reflectance is shown in Fig. 3.

3.2. Analysis of preprocessing method

To optimize the PCE prediction model, the reflectance spectra of the TCA triplelayer were preprocessed by four algorithms SG, STD, D1 and D2 so that different features in the spectral data could be highlighted, and the results are shown in Fig. 4.

Fig. 4(a) demonstrated that the reflectance spectra processed by SG algorithm show distinct peaks around approximately 610 nm and 810 nm and troughs near 500 nm, which was same with that of the original spectra (see Fig. 2(a)). This happens because SG is a local polynomial regression filter that smooths data by fitting a polynomial without changing the position of peaks and valleys on the original spectrum. As far as the reflectance spectra preprocessed by STD was concerned, Fig. 4(b) demonstrated that distribution of the spectra was narrowed for the wavelengths larger than 600 nm, which is attributed to the fact that STD converts spectral data into a standard normal distribution with a mean of 0 and a standard deviation of 1.

In Fig. 4(c), D1 algorithm performs a derivative operation on the spectral data and the resulted spectra are more emphasized on the change rate of the spectral data rather than the value of the spectral data. For example, the reflectance spectra processed by D1 method showed peaks near 600 nm and valleys near 650 nm where original reflectance spectra vary fast (see Fig. 4(a)). Fig. 4(d) showed the second derivative of the spectra data when D2 algorithm was performed on the original spectra. It was found in this figure that D2 algorithm can amplify the small fluctuations in the original spectral data. For example, reflectance spectra processed by D2 algorithm had a valley around 620 nm and multiple feature peaks in the range of 400–500 nm where the original spectra change steadily (see Figs. 2(a) and Fig. 4(a)). These indicated that the D2 method can make the subtle spectral differences more apparent.

In conclusion, different preprocessing method are able to highlight different features in the spectral data based on their unique principles, thus exerting different impacts on the performance of the ML model (See Appendix B for details).

3.3. Feature extraction analysis

3.3.1. Feature extraction analysis of PCA

The PCA algorithm was used to extract the principal components (PCs) which can represent most of the information of the original data. Using the spectral data preprocessed by STD as an example, the PCA algorithm selected 11 PCs of which the cumulative contribution rate reached 95%. The contribution rate of each selected PC was plotted in Figure A1 of Appendix A and the first six PCs have contribution rates of 42.48%, 18.95%, 15.54%, 6.51%, 5.37%, and 4.49%, respectively. To gain a deeper understanding about the importance of the PCs extracted by PCA, a loading plot was used to examine the relationship between the six PC and the original variables, as shown in Fig. 5(a). The results show that the load weight plot shows significant peaks or valleys at 400 nm, 477 nm, 571 nm, 607 nm, 703 nm, and 820 nm, which play an important role in the selected PCs. These peaks or valleys are most strongly correlated with the absorption peaks of the active layer within the OSC device, which are most strongly related to the PCE of OSCs. For example, the features occurring at approximately 477 nm in PC2 and 703 nm in PC5 are likely to be associated with the absorption peaks for OSC1, 4 and 6. The peaks at 571 nm in PC4 may be associated with the absorption peak of the PTB7: PCBM-based OSC device. The peaks at 607 nm in PC3 may correspond to the absorption valley observed for OSC 1, 4, 5 and 6. Furthermore, the features occurring around 820 nm in PC1 may also correspond to the absorption valley observed in OSC 1, 2, 4 and 6.

Further analysis on the relationship between these peak/valley wavelengths and PCE was performed by calculating the distance correlation coefficient and the results are shown in Fig. 5(b) [49,50]. It was found that the distance correlation coefficients at the wavelengths of 703 nm, 571 nm, 477 nm, 607 nm and 820 nm were found to be 0.749, 0.726, 0.712, 0.655 and 0.635, which indicated that strong correlation

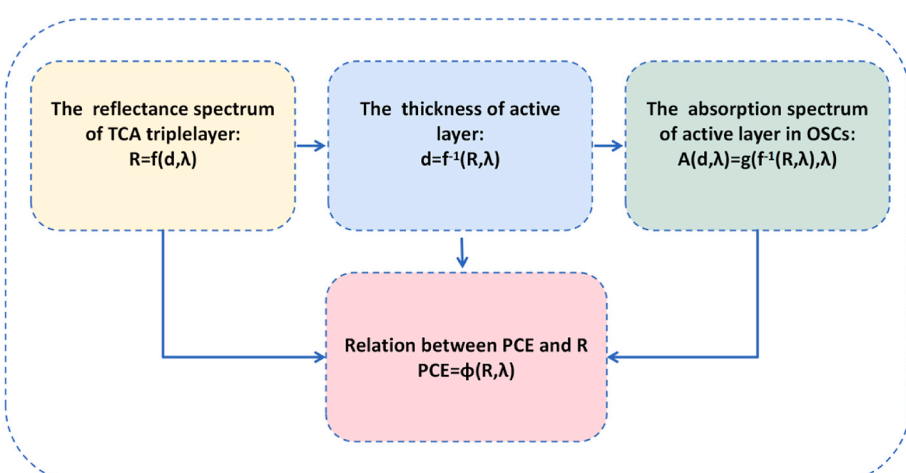

Fig. 3. Schematic diagram of the process from the reflectance of TCA triplelayer to PCE prediction.

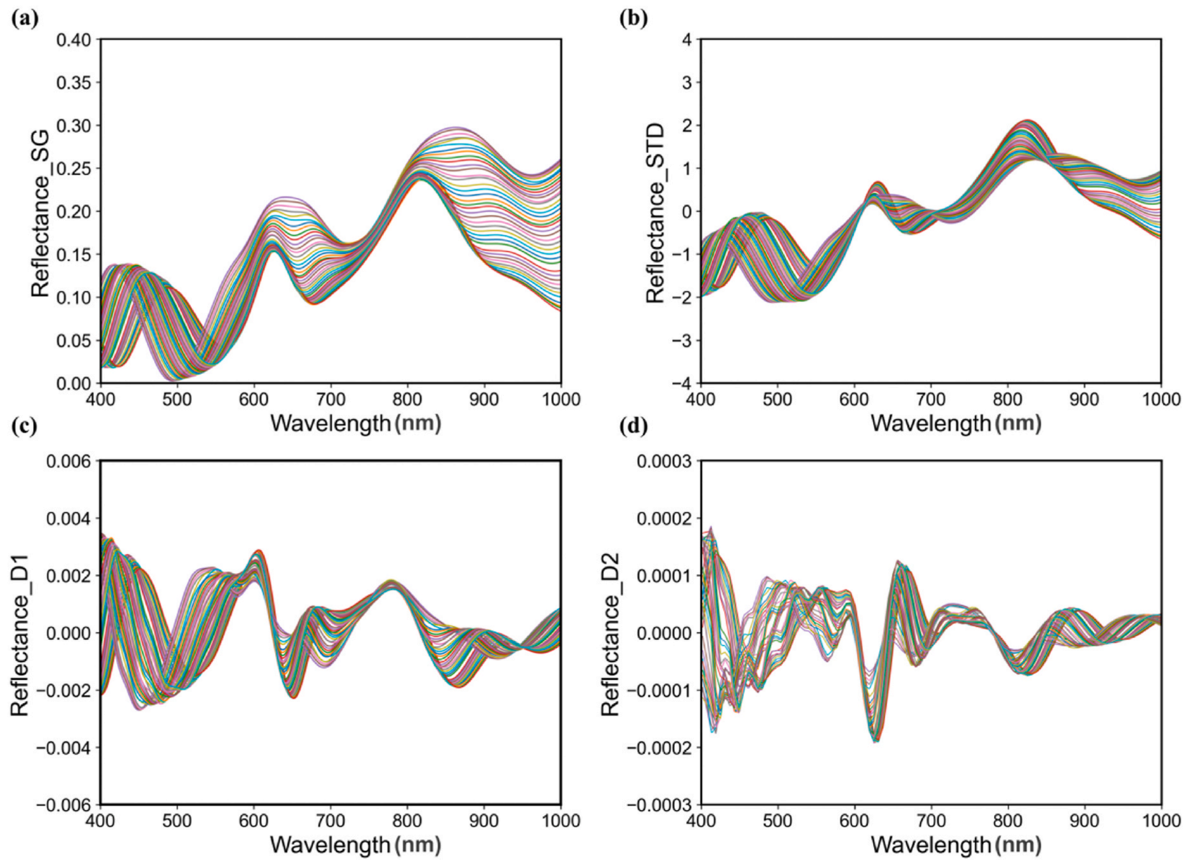

Fig. 4. Comparison of pretreatment methods: (a)SG (b)STD (c)D1 (d)D2.

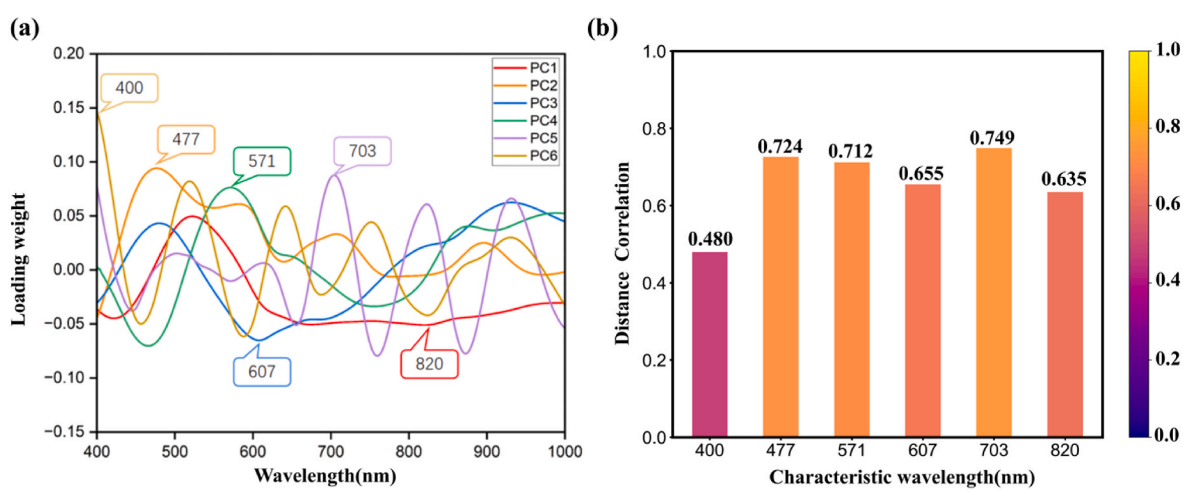

Fig. 5. Illustration of features extracted by PCA. (a) loading weight of wavelengths using the PCA method. (b) the distance correlation between PCE and and the key peak/valley wavelengths of load weight plot of PCs.

occurred between the PCE and theses wavelengths [51,52]. In addition, the correlation coefficients at 400 nm reached a relatively high value of 0.480, indicating a moderate correlation [51,52]. These findings revealed that there were a strong and complex relationship between the reflectance of TCA triplelayer and the PCE of OSC devices, and the important features related to PCE prediction could be extracted by the PCA method.

3.3.2. *Feature extraction analysis of CARS*

In order to remove redundant features in the spectrum and improve the accuracy of the model, the CARS algorithm was also employed to select the characteristic wavelengths of the spectral data, where the Monte Carlo iterations were set to 100 times, utilizing five-fold cross-validation. The final set of characteristic wavelengths was determined by selecting the subset of wavelengths that yielded the lowest $RMSE_{cv}$. The characteristic wavelengths extracted by CARS were shown in Fig. 6 (a), and the variations in parameters associated with the extraction process are detailed in Figure A.2 of Appendix A. **The characteristic wavelengths extracted by CARS are shown in** Fig. 6(a), and the results show that the characteristic wavelengths are mainly concentrated

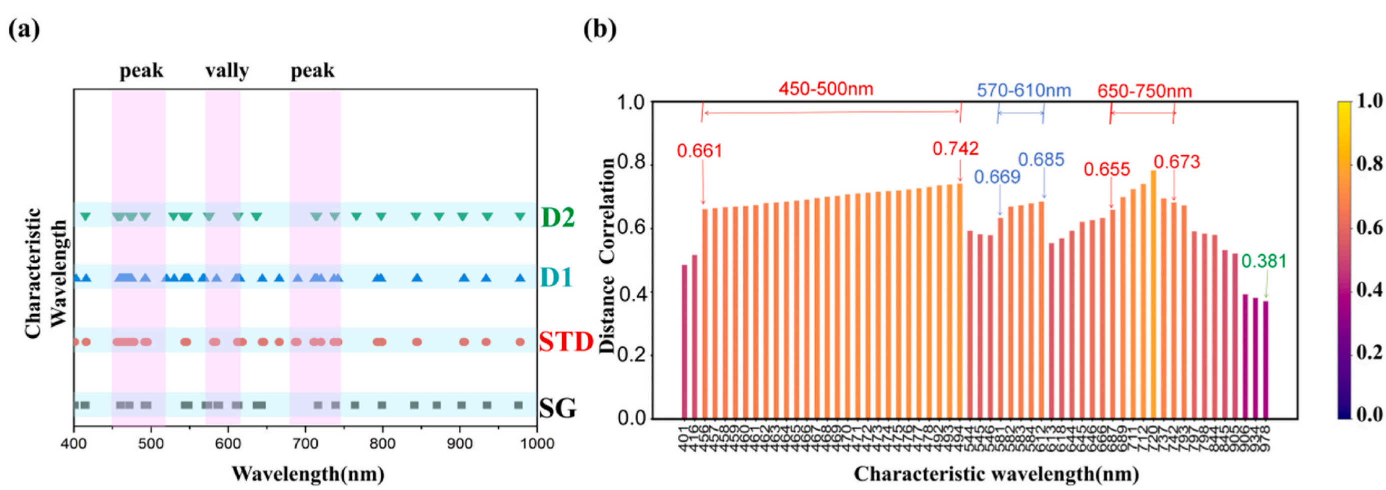

Fig. 6. Illustration of features extracted by CARS. (a) characteristic wavelength extracted by CARS method. (b) the correlation between wavelengths extracted by CARS and PCE.

in 450 nm–500 nm, 570 nm–630 nm, and 650 nm–750 nm, which are mainly near the peaks and valleys of the OSC absorption spectrum. For example, the absorption spectrum of the corresponding OSC 2 with PTB7: PCBM as the active layer exhibits dips at 470 nm and 663 nm, with a peak at 583 nm, respectively (see Fig. 2(d)). Further distance correlation coefficients are shown in Fig. 6(b). The correlation coefficient between PCE and the characteristic wavelengths in the range of 450 nm–500 nm, 570 nm–610 nm, and 650 nm–750 nm exceeds 0.6, indicating that these wavelengths have a strong correlation with PCE [51,52]. In addition, the correlation coefficients between the other range of characteristic wavelengths and the PCE is over 0.38, indicating that these wavelengths have a moderate correlation with PCE [51,52].

3.4. Performance analysis of different models

When training the model, 70% of the dataset (424 samples) is randomly allocated to the training set, while the remaining 30% (182 samples) is allocated for the test set. To verify the effectiveness of the sample division between the test set and the training set, the statistical distribution, mean, and median values of the PCE for the training set and the test set were calculated. The results are presented in Fig. 7. It can be seen that there is little difference in the distribution of PCE between the training and test sets. Furthermore, the mean and median PCE values for the training set are 11.55 and 10.7, respectively, which is comparable to that (11.53 and 10.33) for the test set. From a statistical perspective, the test set and the training set exhibit comparable data characteristics, suggesting that the division process is fair and there is no bias due to random partitioning.

To construct the optimal ML model for predicting the PCE of OSCs based on the reflectance spectra of the TCA triplelayer, the original spectral data are initially subjected to four different pre-processing techniques (SG, STD, D1, D2) to improve data quality. Subsequently, two feature extraction methods, PCA and CARS, are employed to reduce the dimensionality of the data and extract the important features. Finally, the extracted features from the eight treatment combinations (SG-PCA, SG-CARS, STD-PCA, STD-CARS, D1-CARS, D1-PCA, D2-CARS, D2-PCA) are individually fed into six ML models, resulting in a total of 48 regression models. The performance of these models is shown in Appendix B, while the performance of the optimized models is compared in Table 3, Tables 4 and 5. Fig. 7 shows the scatter plots of the models of D2-PCA-PLSR, STD-PCA-RFR, STD-PCA-KR and STD-PCA-MLPR. Two principal conclusions can be drawn from these results.

Firstly, for the models trained on the data without pre-processing and feature extraction, the R^2_{test} for PLSR, KR, RFR, SVR, ELM and MLPR are 0.506, 0.835, 0.849, 0.864, 0.794 and 0.904, respectively (see Table 3). These values are much lower than those of the models optimized by the pre-processing and feature extraction method, as can be seen in Tables 4 and 5. Furthermore, the $RMSE_{test}$ for these models is also reduced when the data is treated by the pre-processing and feature extraction method. Compared with the models without preprocessing and feature extraction, the R^2_{test} of the six models, PLSR, KR, RFR, SVR, ELM and MLPR, increased by 0.286, 0.144, 0.125, 0.08, 0.133 and 0.08 respectively after optimization. This happens because the pre-processing method serves to enhance the quality and reliability of the reflectance spectra, while the feature extraction method allows for the efficiently removal of irrelevant information from the reflectance spectra [53,54]. Particularly, the MLPR model processed with STD-PCA stands out. The STD eliminates the differences between reflectance data of the six types of TCA triplelayer, making the data more comparable [32]. After preprocessing the original data, PCA further extracts the main features that best represent the variability of the data, thereby improving the prediction accuracy of the model [55].

Secondly, the MLPR model combined with PCA achieves the highest R^2_{test} value of 0.984, which is considerably higher than that of the PLSR model (value of 0.792). This can be attributed to the intricate and non-linear relationships between the absorption of the active layer within the OSC devices and the reflectance spectra of TCA triplelayer, which is due to the optical interference within the device (see Equation (6)). Therefore, MLPR is a more suitable technique than PLSR for constructing a prediction model for PCE, as it is capable of learning more complex non-linear relationships due to two key factors [56–58]. On the one hand, MLPR contains multiple hidden layer neurons, which enables each layer of neurons to learn different abstract features of the data. For example, the neurons in the first layer may discern fundamental waveform features in the reflectance spectrum, whereas subsequent layers can combine these basic features to construct more high-orderly PCE-related features. The multi-layered structure of MLPR enable itself to automatically identify non-linear hierarchies in data. In contrast, PLSR processes data mainly based on linear combinations, which hinders the extraction of intricate non-linear relationships. On the other hand, MLPR employs a backpropagation algorithm to adjust the weights of the connections between neurons. When a minor change in the reflectance spectrum in a certain wavelength range has a non-linear impact on the PCE, the backpropagation algorithm can adjust the weights so that the model can accurately learn the pattern of change. In comparison, the coefficient update method of PLSR is relatively fixed and mainly based on the principle of linear fitting, which limits its capacity to learn nonlinear relations.

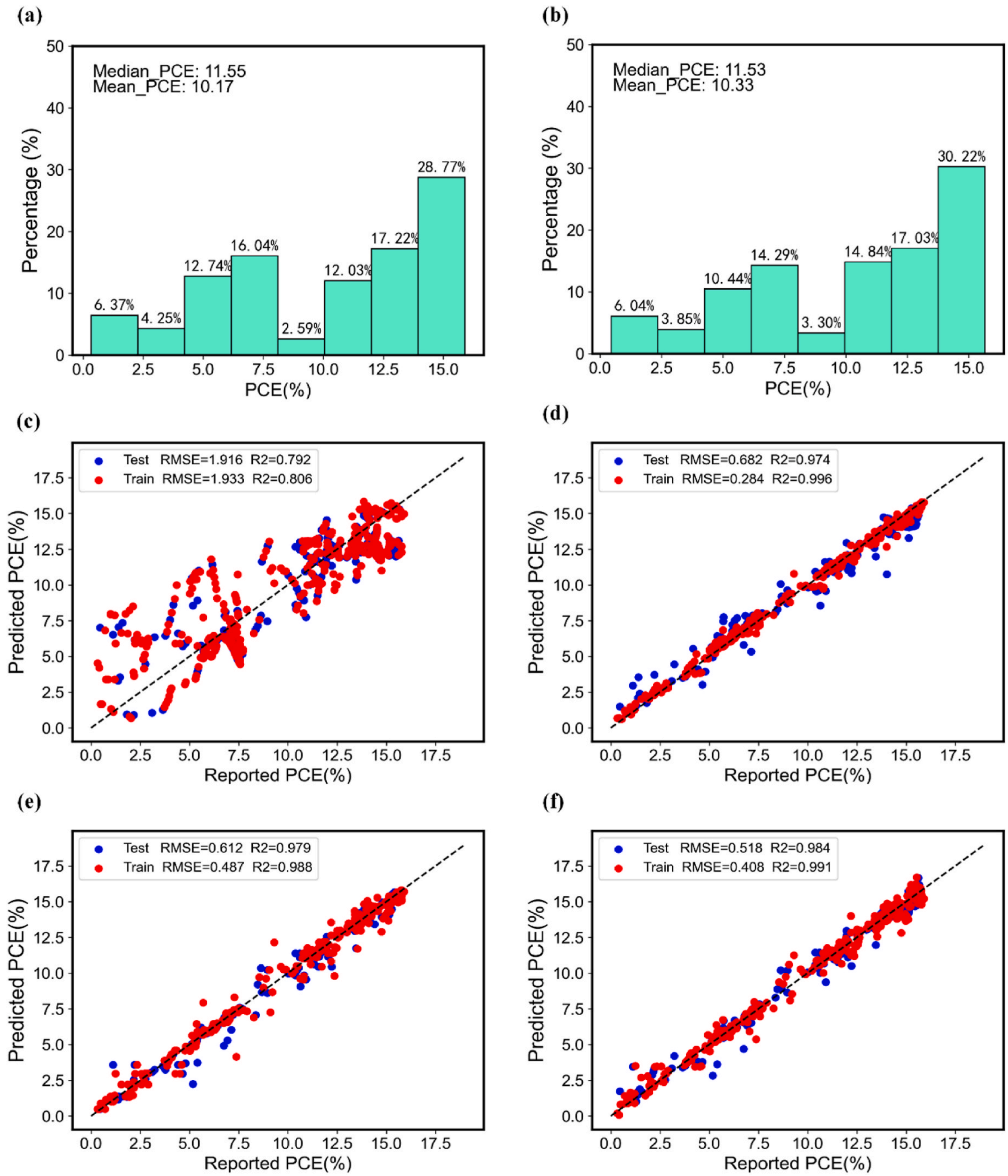

Fig. 7. Distribution of PCE for samples in the training set (a) and in test set (b). Scatter plots of the models of (c) D2-PCA-PLSR, (d)STD-PCA-RFR, (e)STD-PCA-KR and (f)STD-PCA-MLPR.

Table 3
Performance of optimized model without pre-processing and feature extracting.

Model	R^2_{train}	$RMSE_{tarin}$	R^2_{test}	$RMSE_{test}$
PLSR	0.528	3.011	0.506	2.954
KR	0.883	1.479	0.835	1.766
RFR	0.895	1.363	0.849	1.634
SVR	0.891	1.447	0.864	1.551
ELM	0.830	1.806	0.794	1.909
MLPR	**0.943**	**1.051**	**0.904**	**1.302**

Table 4
Performance of optimized model with features extracted by PCA.

Method	Model	R^2_{train}	$RMSE_{tarin}$	R^2_{test}	$RMSE_{test}$
D2	PLSR	0.806	1.933	0.792	1.916
STD	KR	0.988	0.487	0.979	0.612
STD	RFR	0.996	0.284	0.974	0.682
STD	SVR	0.954	0.940	0.944	0.993
D1	ELM	0.943	1.043	0.927	1.138
STD	**MLPR**	**0.991**	**0.408**	**0.984**	**0.518**

Table 5
Performance of optimized model with features extracted by CARS.

Method	Model	R^2_{train}	$RMSE_{tarin}$	R^2_{test}	$RMSE_{test}$
STD	PLSR	0.579	2.845	0.561	2.780
STD	KR	0.966	0.803	0.949	0.947
STD	RFR	0.995	0.312	0.962	0.814
D1	SVR	0.943	1.043	0.940	1.030
D1	ELM	0.915	1.282	0.867	1.532
STD	**MLPR**	**0.991**	**0.426**	**0.981**	**0.607**

3.5. Learning curve analysis

A learning curve analysis is performed to ascertain whether the models (KR, RFR, MLPR) are overfitting or not. For this purpose, the models were trained with an increasing number of training samples, ranging from 50 to 350. The resulting RMSE values for both the training and test sets are shown in Fig. 8. Initially, with only 50 samples used for training, the RMSE for the test set was significantly higher than that for the training set, indicating that all models were overfitting. As the number of samples increased to 350, the RMSE for the test set rapidly decreased to a stable value, with a slight elevation compared to the training set. These results confirm that the models do not exhibit overfitting when trained with a large number of samples. More importantly, the RMSE of the MLPR reached a minimal value of 0.668, which is much lower than that of the other models. This finding supports the conclusion that the MLPR exhibits a better predictive accuracy than other models. In order to further explore the generalization ability and stability of MLPR with STD-PCA, K-fold cross-validation was used in this study, and the results were placed in Appendix C. The average values of $R^2$and RMSE obtained by K-fold cross validation are 0.982 and 0.547, respectively, which are close to the previously obtained values of 0.984 and 0.518. The results show that the MLPR model shows consistency and stability in different data divisions.

3.6. Potential application of the model

Up to this, the investigation demonstrated that the ML models can be used to predict the PCE of the OSC device from the reflectance of the TCA triplelayer by assuming that all generated photoelectrons could contribute to current and setting fill factor and open-circuit voltage as empirical values derived from published literatures. In practical device fabrication, however, the J_{sc}, V_{oc} and FF, are determined by several other factors such as carrier transport and collection, morphology of the active layer, interface matching, etc. In other word, these practical factors should be taken into consideration to construct applicable PCE prediction models. The issue can be addressed by integrating hyperspectral camera into the OSC fabrication equipment to measure the reflectance spectra of the TCA triplelayer, as shown in Fig. 9. In the experimental setups, the hyperspectral camera is positioned directly above the spin-coating platform, which is illuminated by halogens. In the fabrication process of the OSCs, the PCE prediction models can be constructed by the following steps: (1) employ the hyperspectral camera to capture the reflectance spectra of the TCA triplelayer after the active layer has been spin-coated; (2) obtain the PCE of the OSC devices by measuring the J-V curves after the device has been fabricated; (3) train the PCE prediction models based on the measured reflectance spectra and PCE.

The ML-based PCE prediction models based on the measured reflectance spectra and PCE, together with a hyperspectral camera and

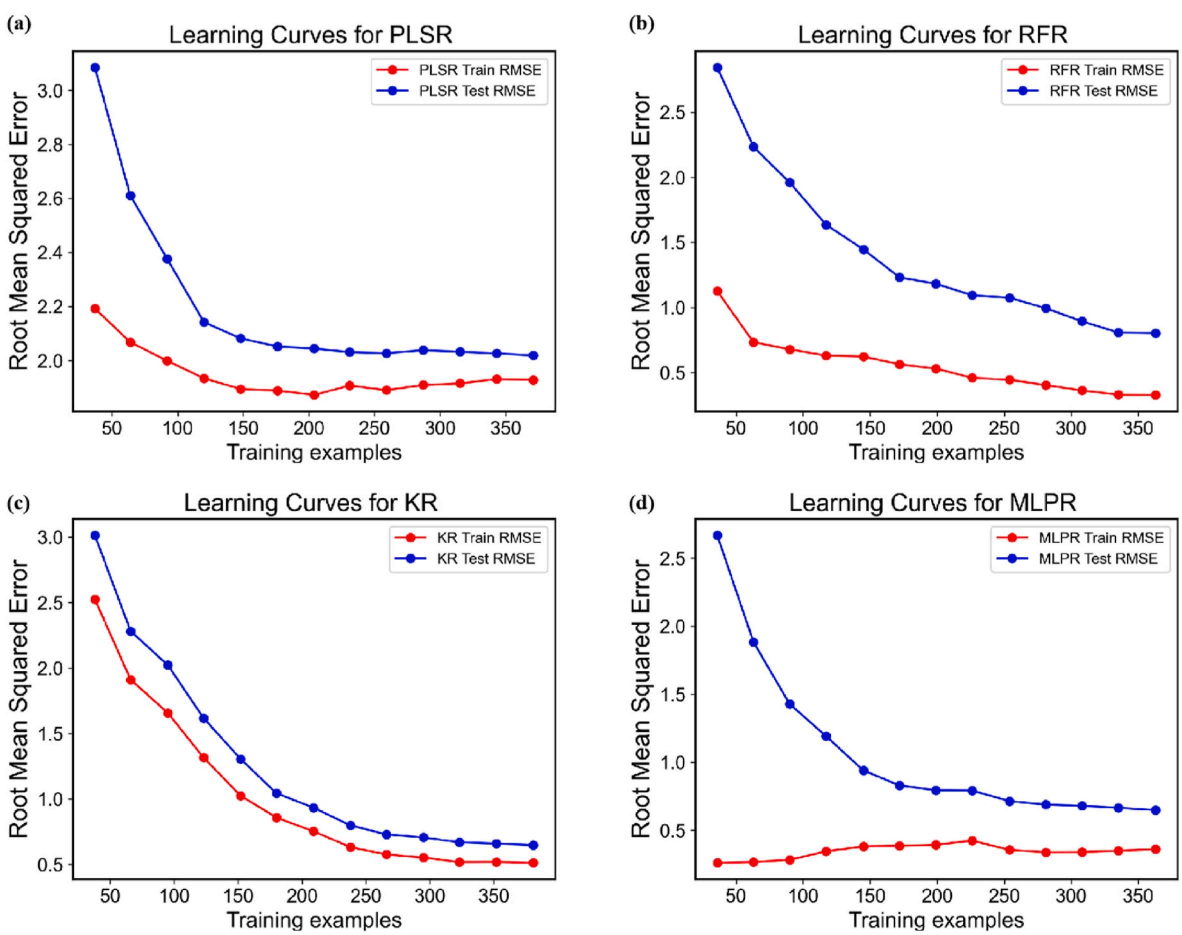

Fig. 8. Learning curve for (a) D2-PCA-PLSR (b) STD-PCA-RFR (c) STD-PCA-KR (d) STD-PCA-MLPR.

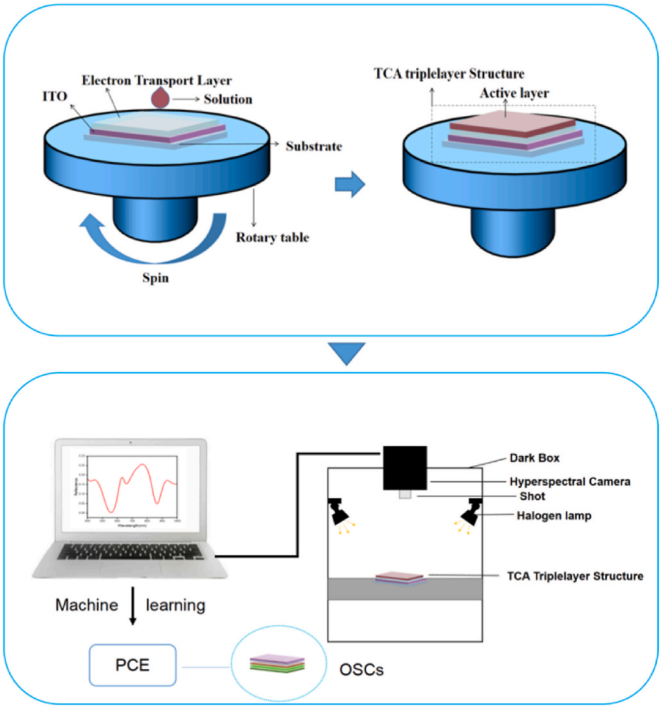

Fig. 9. The proposed application scenarios of the constructed ML regression model.

computer, can be integrated into the fabrication equipment to monitor the quality of the TCA triplelayer, as shown in Fig. 9. In the experimental setups, the hyperspectral camera is used to capture the reflectance spectra of the TCA triplelayer and the captured reflectance data is then uploaded to the computer for further analysis. The ML regression model, which is integrated into a spectral analysis software system on the computer, can be used to predict the PCE of the OSCs based on the TCA triplelayer. Based on the predicted PCE, the quality of the TCA triplelayer can be evaluated and a decision made as to the remaining fabrication process should be completed. The proposed method allows for the effective discrimination of TCA triplelayer with suboptimal performance, obviating the necessity for subsequent manufacturing processes for these samples. This will further enhance the preparation yield and reduce manufacturing costs. Therefore, a novel perspective can be introduced for evaluating the quality of device fabrication processing.

In practical research, hyperspectral images can provide both spatial and spectral information of the TCA triplelayer, which contains the morphology of the active layer, the energy band (LUMO level, HOMO level) of organic materials, and the interactions between organic molecules in the active layer [59–61]. It indicates that hyperspectral images may bring information related to the practical J_{SC}, FF and V_{oc} of OSCs. In other word, hyperspectral imaging combined with ML models may be used to predict the practical J_{sc}, FF and V_{oc} of the OSCs, and then used to predict the PCE of the device.

Nevertheless, the investigation in this paper demonstrates that the PCE of the device can be predicted from the calculated reflectance spectra of the TCA triplelayer when only optical factors are taken into consideration. In future research, more experiments will be conducted to construct a practical ML regression model involving practical factors such as carrier transport and collection, morphology of the active layer, interface matching, to predict the PCE of the OSC device from the measured hyperspectral images of the TCA triplelayer.

Another point should be pointed out is that the prediction models in this paper are constructed on bulk heterojunction OSCs with five kinds of active layers, which is composed by four kinds of polymer donors (PM6, PTB7, P3HT, PBDB-T), two kinds of small molecule acceptor (Y6, ITIC) and three kinds of polymer acceptor ($PC_{71}BM$, PCBM, BTP-ec9). Further experiments should be performed to examine the applicability of the approach in both bulk heterojunction and bilayer heterojunction OSCs with other material systems including various small molecule donor and acceptor, polymer donor and acceptor, and oligomer.

4. Conclusions

Machine learning models are constructed to predict the PCE of OSCs based on the reflectance spectra of the TCA triplelayer. This approach aims to provide new methods for optimizing the current experimental procedures and estimating the efficiency of OSCs. For this purpose, a dataset comprising PCEs of six types of OSCs with different active layer materials and reflectance spectra of TCA triplelayer was firstly constructed by the FDTD method. Subsequently, machine learning algorithms were employed to construct the PCE prediction models. By comparing different spectrum pre-processing and feature extraction methods, we analyzed the correlation between the reflectance spectra of the TCA triplelayer and the efficiency of the OSC, and optimized the prediction accuracy of the regression model. The results showed that the MLPR algorithm using STD-PCA pre-processing achieved the best performance, with an R^2 of 0.984 and a relatively low RMSE of 0.408 among the 48 regression results. This indicates that the PCE of OSCs can be accurately predicted based on the reflectance spectra of the TCA. Furthermore, this paper proposes a new perspective based on the constructed regression model to monitor the quality of the three layers of TCA in the device manufacturing process, offering a rapid and non-destructive method to evaluate important optoelectronic characteristics of OSCs.

CRediT authorship contribution statement

Fuhao Gao: Writing – original draft, Software, Investigation, Data curation, Conceptualization. **Jinxin Zhou:** Visualization, Validation. **Junwei Zhao:** Software, Data curation. **Senxuan Lin:** Data curation. **Jingfeng Liu:** Writing – review & editing. **Yubin Lan:** Funding acquisition. **Yongbing Long:** Writing – review & editing, Supervision, Methodology, Funding acquisition. **Haitao Xu:** Writing – review & editing, Validation, Supervision, Funding acquisition, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

The authors are grateful to National Natural Science Foundation of China (12174120, 11774099), The 111 Project (D18019), Leading talents of Guangdong province program (2016LJ06G689) and Laboratory of Lingnan Modern Agriculture Project(NT2021009)for the support to the work.

Appendix A. Feature extraction

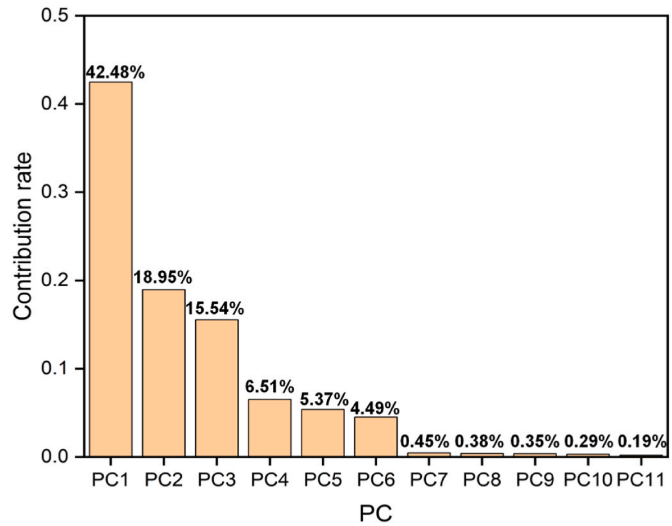

Fig. A.1. Contribution rate of principal components extracted by PCA

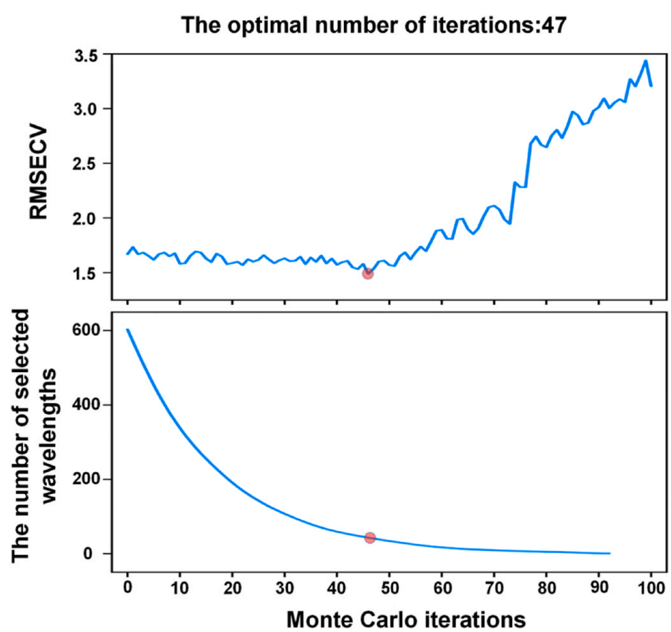

Fig. A.2. the change of relevant parameters during feature wavelength extraction by CARS.

Table A.1
The optimal number of dimensions for CARS and Characteristic Wavelength

method	number	Characteristic Wavelength(nm)
SG	45	401,414,415,460,461,472–477,492~495,544–548,571~574,586–588,610~613,637–639,643,716,739,764,799,841,870,902,935,976
STD	52	401,416,456–478,492~494,544–546,581~584,612–615,618,644–646,666,687,689,711,712,720,737,742,793,797,798,844,845,905,906,934,978
D1	55	403,416,462–479,492,493,520,530,543–549,567~569,585,610,612,614,615,644,666,690,712,713,715,720,737,742,793,797,798,844,845,905,906,934,978
D2	40	407,456–461,474~478,492,493,529,542–547,574~576, 612, 613, 636, 637, 713, 715, 738, 766, 798, 843, 873, 903, 904, 935, 936, 978

Appendix B. The performance of each model using PCA and CARS

Table B.1
The prediction performance of each model using PCA.

Method	Model	R^2_{train}	$RMSE_{tarin}$	R^2_{test}	$RMSE_{test}$
STD	PLSR	0.760	2.147	0.750	2.100
SG		0.758	2.156	0.746	2.118
D1		0.789	2.015	0.768	2.024
D2		0.806	1.933	0.792	1.916
STD	KR	0.988	0.487	0.979	0.612
SG		0.988	0.514	0.973	0.689
D1		0.985	0.543	0.952	0.721
D2		0.983	0.573	0.974	0.677
STD	RFR	0.996	0.284	0.974	0.682
SG		0.996	0.287	0.970	0.723
D1		0.994	0.352	0.952	0.921
D2		0.994	0.345	0.970	0.712
STD	SVR	0.954	0.940	0.944	0.993
SG		0.946	1.017	0.930	1.109
D1		0.943	1.043	0.940	1.030
D2		0.907	1.284	0.905	1.202
STD	ELM	0.897	1.407	0.824	1.764
SG		0.833	1.717	0.819	1.782
D1		0.943	1.043	0.927	1.138
D2		0.841	1.680	0.836	1.753
STD	MLPR	0.991	0.408	0.984	0.518
SG		0.989	0.462	0.979	0.609
D1		0.949	0.788	0.934	0.793
D2		0.975	0.548	0.946	0.701

Table B.2
The prediction performance of each model using CARS.

Method	Model	R^2_{train}	$RMSE_{tarin}$	R^2_{test}	$RMSE_{test}$
STD	PLSR	0.579	2.845	0.561	2.780
SG		0.564	2.785	0.555	2.927
D1		0.514	3.058	0.508	2.947
D2		0.566	2.889	0.560	2.820
STD	KR	0.966	0.803	0.949	0.947
SG		0.951	0.971	0.947	0.963
D1		0.958	0.958	0.936	1.066
D2		0.950	0.985	0.938	1.050
STD	RFR	0.995	0.312	0.962	0.814
SG		0.995	0.305	0.961	0.826
D1		0.995	0.285	0.955	0.887
D2		0.993	0.371	0.962	0.842
STD	SVR	0.891	1.390	0.883	1.435
SG		0.904	1.304	0.899	1.336
D1		0.943	1.043	0.940	1.130
D2		0.885	1.473	0.877	1.490
STD	ELM	0.816	1.880	0.796	1.899
SG		0.878	1.520	0.769	2.018
D1		0.915	1.282	0.867	1.532
D2		0.884	1.495	0.852	1.617
STD	MLPR	0.991	0.426	0.981	0.607
SG		0.945	0.805	0.934	0.897
D1		0.929	0.901	0.924	0.972
D2		0.932	0.957	0.927	0.994

Appendix C. The Performance of MLPR with STD-PCA for 10-fold cross-validation

Table C.1
Performance of MLPR with STD-PCA for 10-fold cross-validation.

K-fold	R^2_{test}	$RMSE_{test}$
2	0.979	0.613
3	0.966	0.794
4	0.978	0.652
5	0.988	0.467
6	0.986	0.478
7	0.986	0.499
8	0.990	0.431
9	0.987	0.500
10	0.986	0.488
Mean	0.982	0.547

Data availability

Data will be made available on request.

References

[1] L. Zhang, Y. He, W. Deng, X. Guo, Z. Bi, J. Zeng, H. Huang, G. Zhang, C. Xie, Y. Zhang, X. Hu, W. Ma, Y. Yuan, X. Yuan, High-efficiency flexible organic solar cells with a polymer-incorporated pseudo-planar heterojunction, Discov. Nano. 19 (1) (2024) 39, 10.1186/s11671-024-03982-1.

[2] R. Lin, H. Zhou, X. Xu, X. Ouyang, Improved charge transport based on donor-acceptor type solid additive with large dipole moment for efficient organic solar cells, Dyes Pigments 224 (2024) 111980, 10.2139/ssrn.4680318.

[3] R. Deka, D.J. Kalita, Boosting the performance of diketopyrrolopyrrole-triphenylamine-based organic solar cells via π-linker engineering, J. Phys. Chem. A 128 (10) (2024) 1753–1766, 10.1021/acs.jpca.3c06439.

[4] S. Guan, Y. Li, C. Xu, N. Yin, C. Xu, C. Wang, M. Wang, Y. Xu, Q. Chen, D. Wang, L. Zuo, H. Chen, Self-assembled interlayer enables high-performance organic photovoltaics with power conversion efficiency exceeding 20, Adv. Mater. 36 (25) (2024) e2400342, 10.1002/adma.202400342.

[5] B. Talukdar, S. Buragohain, S. Kumar, V. Umakanth, N. Sarmah, S. Mahapatra, Effect of spectral response of solar cells on the module output when individual cells are shaded, Sol. Energy 137 (2016) 303–307, 10.1016/j.solener.2016.08.032.

[6] F. Behrouznejad, X. Li, A.A. Umar, X. Zhang, R. Khosroshahi, S.K. Md Saad, I. Ahmed, N. Taghavinia, Y. Zhan, The fingerprint of charge transport mechanisms on the incident photon-to-current conversion efficiency spectra of perovskite solar cells, Sol. Energy Mater. Sol. Cell. 253 (2023) 112234, 10.1016/j.solmat.**2023**.112234.

[7] A. Mahmood, J. Wang, A time and resource efficient machine learning assisted design of non-fullerene small molecule acceptors for P3HT-based organic solar cells and green solvent selection, J. Mater. Chem. A 9 (28) (2021) 15684–15695, 10.1039/d1ta04742f.

[8] A. Mahmood, Y. Sandali, J.L. Wang, Easy and fast prediction of green solvents for small molecule donor-based organic solar cells through machine learning, Phys. Chem. Chem. Phys. 25 (15) (2023) 10417–10426, 10.1039/d3cp00177f.

[9] A. Irfan, M. Hussien, M.Y. Mehboob, A. Ahmad, M.R.S.A. Janjua, Learning from fullerenes and predicting for Y6: machine learning and high-throughput screening of small molecule donors for organic solar cells, Energy. Technol-Ger. 10 (6) (2022) 202101096, 10.1002/ente.202101096.

[10] D. Huang, K. Wang, Z. Li, H. Zhou, X. Zhao, X. Peng, J. Wu, J. Liang, J. Meng, L. Zhao, A machine learning prediction model for quantitative analyzing the influence of non-radiative voltage loss on non-fullerene organic solar cells, Chem. Eng. J. 475 (2023) 145958, 10.1016/j.cej.2023.145958.

[11] M.R.S.A. Janjua, A. Irfan, M. Hussien, M. Ali, M. Saqib, M. Sulaman, Machine-learning analysis of small-molecule donors for fullerene based organic solar cells, Energy. Technol-Ger. 10 (5) (2022) 202200019, 10.1002/ente.202200019.

[12] P. Malhotra, S. Biswas, G.D. Sharma, Directed message passing neural network for predicting power conversion efficiency in organic solar cells, Acs. Appl. Mater. Inter. 15 (31) (2023) 37741–37747, 10.1021/acsami.3c08068.

[13] Y. Cui, Q. Fan, H. Feng, T. Li, D.Y. Paraschuk, W. Ma, H. Yan, Towards efficient and stable organic solar cells: fixing the morphology problem in block copolymer active layers with synergistic strategies supported by interpretable machine learning, Energy Environ. Sci. 17 (22) (2024) 8954–8965, https://doi.org/10.1039/d4ee03168g.

[14] K. Wang, J. Liang, Z. Li, H. Zhou, C. Nie, J. Deng, Y. Zou, Design of experiments with the support of machine learning for process parameter optimization of all-small-molecule organic solar cells, FlexMat 1 (3) (2024) 234–247, 10.1002/flm2.34.

[15] X. Du, L. Lüer, T. Heumueller, A. Classen, C. Liu, C. Berger, C.J. Brabec, Revealing processing stability landscape of organic solar cells with automated research platforms and machine learning, InfoMat (2024) e12554, https://doi.org/10.1002/inf2.12554.

[16] R.T. Furbank, V. Silva-Perez, J.R. Evans, A.G. Condon, G.M. Estavillo, W. He, Z. He, Wheat physiology predictor: predicting physiological traits in wheat from hyperspectral reflectance measurements using deep learning, Plant Methods 20 (2024), https://doi.org/10.1186/s13007-021-00806-6.

[17] T. Long, X. Tang, C. Liang, B. Wu, B. Huang, Y. Lan, Y. Long, Detecting bioactive compound contents in Dancong tea using VNIR-SWIR hyperspectral imaging and KRR model with a refined feature wavelength method, Food Chem. 460 (2024) 140579, https://doi.org/10.1016/j.foodchem.2024.140579.

[18] Q. Li, W. Wang, C. Ma, Z. Zhu, Detection of physical defects in solar cells by hyperspectral imaging technology, Opt Laser. Technol. 42 (6) (2010) 1010–1013, https://doi.org/10.1016/j.optlastec.2010.01.022.

[19] R. Kerremans, C. Kaiser, W. Li, N. Zarrabi, P. Meredith, A. Armin, The optical constants of solution-processed semiconductors—new challenges with perovskites and non-fullerene acceptors, Adv. Opt. Mater. 8 (16) (2020), 10.1002/adom.202000319.

[20] T. Xu, Y. Luo, S. Wu, B. Deng, S. Chen, Y. Zhong, S. Wang, G. Leveque, R. Bachelot, F. Zhu, High-performance semitransparent organic solar cells: from competing indexes of transparency and efficiency perspectives, Adv. Sci. 9 (26) (2022), 10.1002/advs.202202150.

[21] M.N. Polyanskiy, Refractiveindex.info database of optical constants, Sci. Data 11 (1) (2024) 94, 10.1038/s41597-023-02898-2.

[22] K. Yee, Numerical solution of initial boundary value problems involving Maxwell's equations in isotropic media, IEEE Trans. Antenn. Propag. 14 (3) (1966) 302–307, 10.1109/tap.1966.1138693.

[23] G. Dennler, M.C. Scharber, C.J. Brabec, Polymer-Fullerene bulk-heterojunction solar cells, Adv. Mater. 21 (13) (2009) 1323–1338, 10.1002/adma.200801283.

[24] X. Qian, J. Li, J. Zhang, W. Zhang, W. Yue, Q. Wu, H. Zhang, Y. Wu, W. Wang, Micro-crack detection of solar cell based on adaptive deep features and visual saliency. Sensor, Rev. 40 (4) (2020) 385–396, 10.1108/sr-05-2019-0124.

[25] J. Yuan, Y. Zhang, L. Zhou, G. Zhang, H. Yip, T. Lau, X. Lu, C. Zhu, H. Peng, P. A. Johnson, M. Leclerc, Y. Cao, J. Ulanski, Y. Li, Y. Zou, Single-junction organic solar cell with over 15% efficiency using fused-ring acceptor with electron-deficient core, Joule 3 (4) (2019) 1140–1151, 10.1016/j.joule.2019.01.004.

[26] N. Al-Shekaili, S. Hashim, F.F. Muhammadsharif, M.Z. Al-Abri, K. Sulaiman, M. Y. Yahya, M.R. Ahmad, Enhanced performance of PTB7:PC71BM based organic solar cells by incorporating a nano-layered electron transport of titanium oxide, Ecs. J. Solid. State. Sc. 9 (10) (2020) 105003, 10.1149/2162-8777/abc1c1.

[27] D.C. Tripathi, S.K. Gupta, A. Kumar, S.K. Pathak, A. Garg, Impact of recombination mechanisms on the capacitance-voltage characteristics in bulk heterojunction organic solar cells, Synth. Met. (2023) 1–6, 10.1016/j.synthmet.2023.117419.

[28] Z. Yin, S. Mei, P. Gu, H.-Q. Wang, W. Song, Efficient organic solar cells with superior stability based on PM6:BTP-eC9 blend and AZO/Al cathode, iScience 24 (9) (2021) 103027, 10.1016/j.isci.2021.103027.

[29] Q. Zhang, H. Zhang, Z. Wu, C. Wang, R. Zhang, C. Yang, F. Gao, S. Fabiano, H. Y. Woo, M. Ek, X. Liu, M. Fahlman, Natural product betulin-based insulating polymer filler in organic solar cells, Sol. RRL 6 (9) (2022), 10.1002/solr.202200381.

[30] H. Chen, Q. Song, G. Tang, L. Xu, An optimization strategy for waveband selection in FT-NIR quantitative analysis of corn protein, J. Cereal. Sci. 60 (3) (2014) 595–601, 10.1016/j.jcs.2014.07.009.

[31] Y. Liu, B. Dang, Y. Li, H. Lin, H. Ma, Applications of Savitzky-Golay filter for seismic random noise reduction, Acta Geophys. 64 (1) (2016), 10.1515/acgeo-2015-0062.

[32] Q. Lin, Q. Fang, Y. Feng, A. Song, Z. Hou, H. Chen, H. Yue, N. Chen, Z. Wang, Z. Li, G. Xiao, C. Ken, An optimization strategy for detection of fertile pigeon egg based on NIR spectroscopy analysis. Infrared, Phys. Technol. 132 (2023) 104733, 10.1016/j.infrared.2023.104733.

[33] M. Bagtash, J. Zolgharnein, Removal of brilliant green and malachite green from aqueous solution by a viable magnetic polymeric nanocomposite: simultaneous spectrophotometric determination of 2 dyes by PLS using original and first derivative spectra, J. Chemometr. 32 (7) (2018) e3014, 10.1002/cem.3014.
[34] B. Horn, S. Esslinger, M. Pfister, C. Fauhl-Hassek, J. Riedl, Non-targeted detection of paprika adulteration using mid-infrared spectroscopy and one-class classification – is it data preprocessing that makes the performance? Food Chem. 257 (2018) 112–119, 10.1016/j.foodchem.2018.03.007.
[35] S. Wold, K. Esbensen, P. Geladi, Principal component analysis, Chemometr. Intell. Lab. 2 (1–3) (1987) 37–52, 10.1016/0169-7439(87)80084-9.
[36] H. Li, Y. Liang, Q. Xu, D. Cao, Key wavelengths screening using competitive adaptive reweighted sampling method for multivariate calibration, Anal. Chim. Acta. 648 (1) (2009) 77–84, https://doi.org/10.1016/j.aca.2009.06.046.
[37] Z. Long, Y. Wang, X. Liu, L. Yao, Two-step partial least square regression classifiers in brain-state decoding using functional magnetic resonance imaging, PLoS One 14 (4) (2019) e0214937, 10.1371/journal.pone.0214937.
[38] J. Sim, S. Kim, J. Lee, Prediction of protein solvent accessibility using fuzzy k-nearest neighbor method, Bioinformatics 21 (12) (2005) 2844–2849, 10.1093/bioinformatics/bti423.
[39] J. Song, Bias corrections for Random Forest in regression using residual rotation, J. Korean Surg. Soc. 44 (2) (2015) 321–326, 10.1016/j.jkss.2015.01.003.
[40] X. Peng, Y. Wang, The robust and efficient adaptive normal direction support vector regression, Expert Syst. Appl. 38 (4) (2011) 2998–3008, 10.1016/j.eswa.2010.08.089.
[41] G. Huang, Q. Zhu, C. Siew, Extreme learning machine: theory and applications, Neurocomputing 70 (1–3) (2006) 489–501, 10.1016/j.neucom.2005.12.126.
[42] G. Liu, L. Chen, W. Wang, S. Wang, K. Zhu, Z. Shen, New paradigm for watershed model development by coupling machine learning algorithm and mechanistic model, J. Hydrol. 636 (2024) 131264, 10.1016/j.jhydrol.2024.131264.
[43] J. Ren, J. Zhen, J. Zheo, Optimized design of active layers in organic donor-acceptor solar cells, Acta Phys. Sin. 56 (5) (2007) 2868–2872, 10.7498/aps.56.2868.
[44] X. Nan, W. Lai, J. Peng, H. Wang, B. Chen, H. He, Q. Chen, In situ photoelectric biosensing based on ultranarrowband near-infrared plasmonic hot electron photodetection, Adv. Photonics 6 (2) (2024) 026007, 10.1117/1.ap.6.2.026007.
[45] G. Li, J. Zhong, J. Peng, L. Wang, J. Li, L. He, H. Li, M. Gao, Determination of optical constants and thickness of polymer semiconductor thin film with reflectivity fitting method, Laser & Optoelectronics Progress 53 (4) (2016) 043101, 10.3788/LOP53.043101.
[46] H.A. Macleod, H.A. Macleod, Thin-film Optical Filters, CRC press, 2010, https://doi.org/10.1201/9781420073034.
[47] M. Campoy-Quiles, P.G. Etchegoin, D.D.C. Bradley, On the optical anisotropy of conjugated polymer thin films, Phys. Rev. B Condens. Matter 72 (4) (2005) 045209, https://doi.org/10.1103/physrevb.72.045209.
[48] T. Bonnal, A. Belarouci, R. Orobtchouk, E. Prud'Homme, S. Tadier, G. Foray, How to determine the complex refractive index from infrared reflectance spectroscopy? SN Appl. Sci. 2 (2020) 1–9, https://doi.org/10.1007/s42452-020-03869-7.
[49] G.J. Székely, M.L. Rizzo, N.K. Bakirov, Measuring and testing dependence by correlation of distances (2007) 2769–2794, https://doi.org/10.1214/009053607000000505.
[50] C. Ramos-Carreño, J.L. Torrecilla, dcor: distance correlation and energy statistics in Python, SoftwareX 22 (2023) 101326, https://doi.org/10.1016/j.softx.2023.101326.
[51] X. Liang, Y. Qian, Q. Guo, K. Zheng, A data representation method using distance correlation, Front. Comput. Sci. 19 (1) (2025) 191303, https://doi.org/10.1007/s11704-023-3396-y.
[52] P. Schober, C. Boer, L.A. Schwarte, Correlation coefficients: appropriate use and interpretation, Anesth. Analg. 126 (5) (2018) 1763–1768, https://doi.org/10.1213/ane.0000000000002864.
[53] L.M. Bruce, C.H. Koger, J. Li, Dimensionality reduction of hyperspectral data using discrete wavelet transform feature extraction, IEEE Trans. Geosci. Rem. Sens. 40 (10) (2002) 2331–2338, 10.1109/tgrs.2002.804721.
[54] J. Engel, J. Gerretzen, E. Szymańska, J.J. Jansen, G. Downey, L. Blanchet, L. M. Buydens, Breaking with trends in pre-processing? TrAC, Trends Anal. Chem. 50 (2013) 96–106, 10.1016/j.trac.2013.04.015.
[55] J.R. Beattie, F.W. Esmonde-White, Exploration of principal component analysis: deriving principal component analysis visually using spectra, Appl. Spectrosc. 75 (4) (2021) 361–375, 10.1177/0003702820987847.
[56] P. Mishra, F. Marini, A. Biancolillo, J.M. Roger, Improved prediction of fuel properties with near-infrared spectroscopy using a complementary sequential fusion of scatter correction techniques, Talanta 223 (2021) 121693, 10.1016/j.talanta.2020.121693.
[57] S. Mitra, S.K. Pal, Fuzzy multi-layer Perceptron, inferencing and rule generation, IEEE Trans. Neural Network. 6 (1) (1995) 51–63, 10.1109/72.363450.
[58] G. Cybenko, Approximation by superpositions of a sigmoidal function, Math. Control, Signals, Syst. 2 (1989) 303–314, 10.1007/bf02551274.
[59] D. Wu, D.W. Sun, Advanced applications of hyperspectral imaging technology for food quality and safety analysis and assessment: a review—Part II: applications, Innov. Food Sci. Emerg. Technol. 19 (2013) 15–28, 10.1016/j.ifset.2013.04.016.
[60] J. Wen, D.J. Miller, W. Chen, T. Xu, L. Yu, S.B. Darling, N.J. Zaluzec, Visualization of hierarchical nanodomains in polymer/fullerene bulk heterojunction solar cells, Microsc. Microanal. 20 (5) (2014) 1507–1513, 10.1017/s1431927614001615.
[61] A. Delamarre, L. Lombez, J.F. Guillemoles, Mapping solar cell parameters using hyperspectral imaging. SPIE, Newsroom (2013), 10.1117/2.1201304.004777.